

Image Formation

Fitsum Admasu

Department of Computer Science

Addis Ababa University

Objective

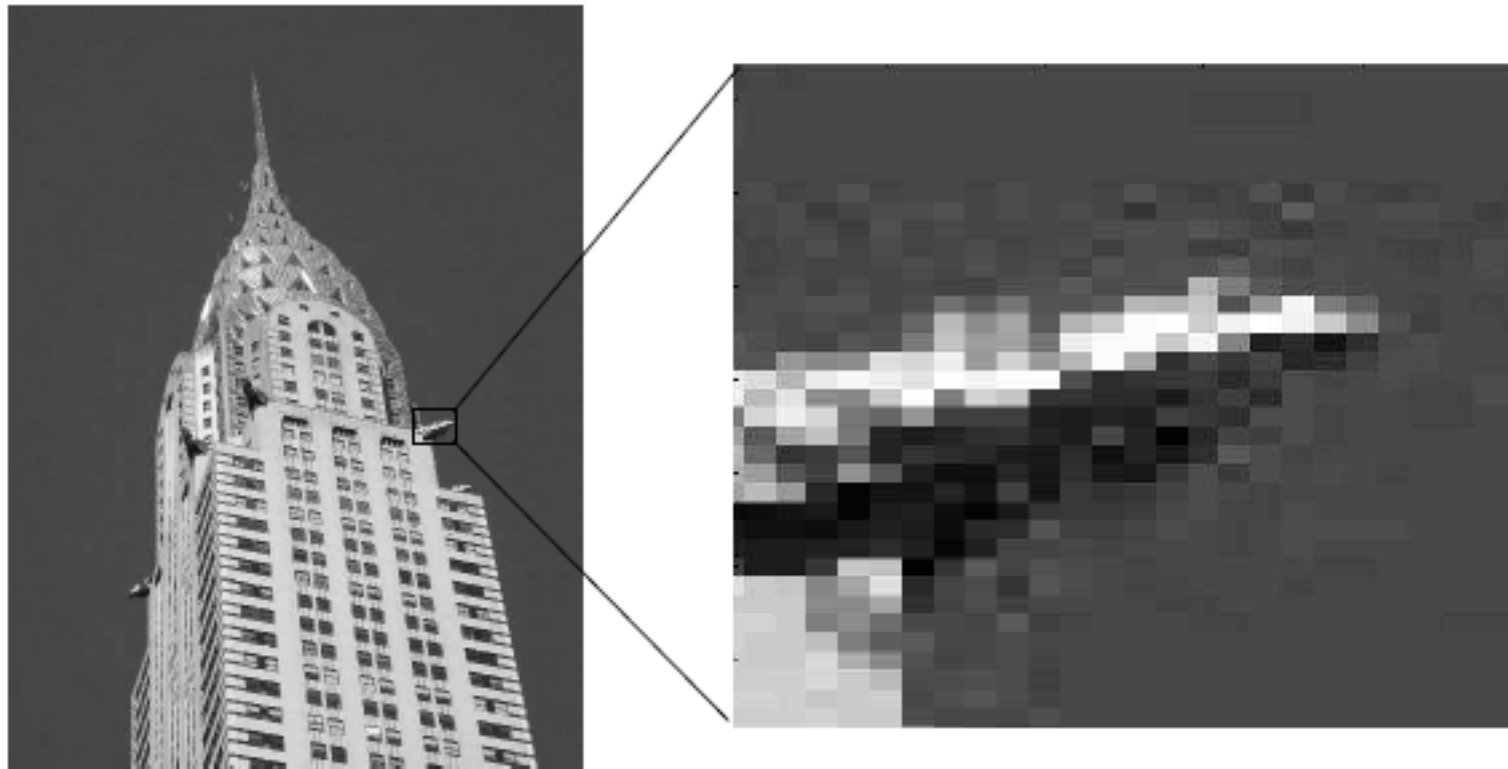
- To provide concepts about digital image representation.
- Understand image quantitative and qualitative descriptions
 - Contrast
 - Resolution
- Digital image acquisition

Gray Level Images

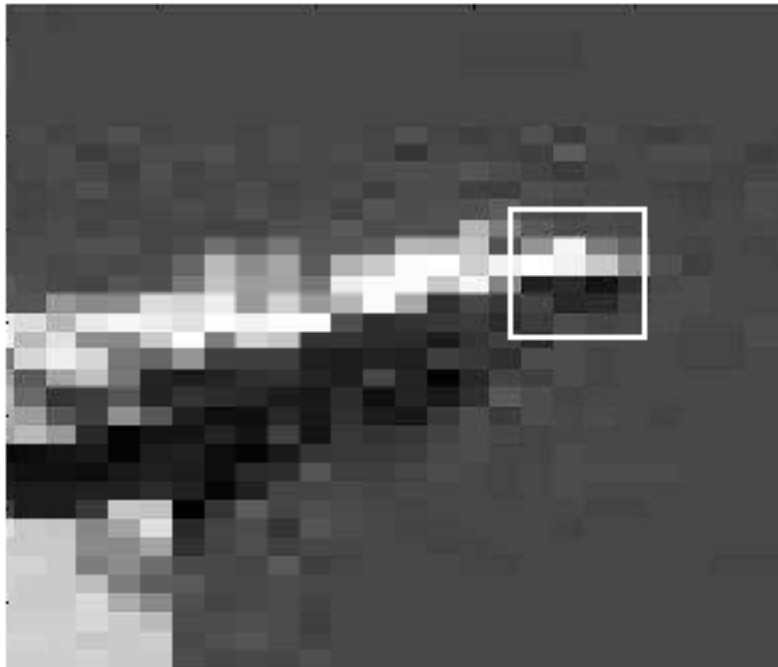


- These are referred to as **grayscale** or **gray level** images
- Corresponds to monochromatic light
 - Light “devoid” of color

Gray Level Images



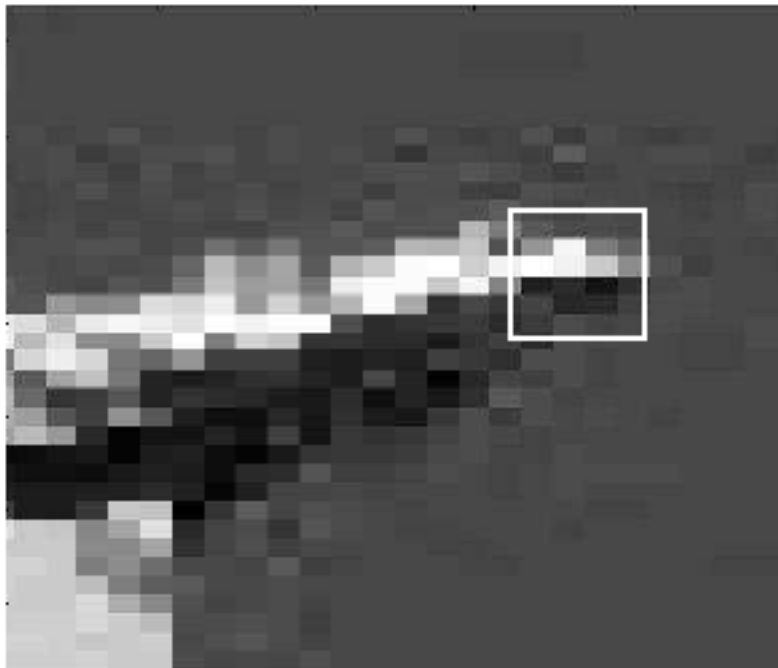
Gray Level Images



<u>61</u>	<u>29</u>	<u>29</u>	<u>57</u>	<u>199</u>	<u>192</u>
<u>222</u>	<u>200</u>	<u>197</u>	<u>135</u>	<u>167</u>	<u>222</u>
<u>203</u>	<u>203</u>	<u>203</u>	<u>137</u>	<u>137</u>	<u>165</u>
<u>208</u>	<u>208</u>	<u>201</u>	<u>124</u>	<u>142</u>	<u>111</u>
<u>208</u>	<u>203</u>	<u>200</u>	<u>190</u>	<u>127</u>	<u>92</u>
<u>204</u>	<u>201</u>	<u>200</u>	<u>218</u>	<u>173</u>	<u>139</u>

It's just a bunch of NUMBERS!

Gray Level Images



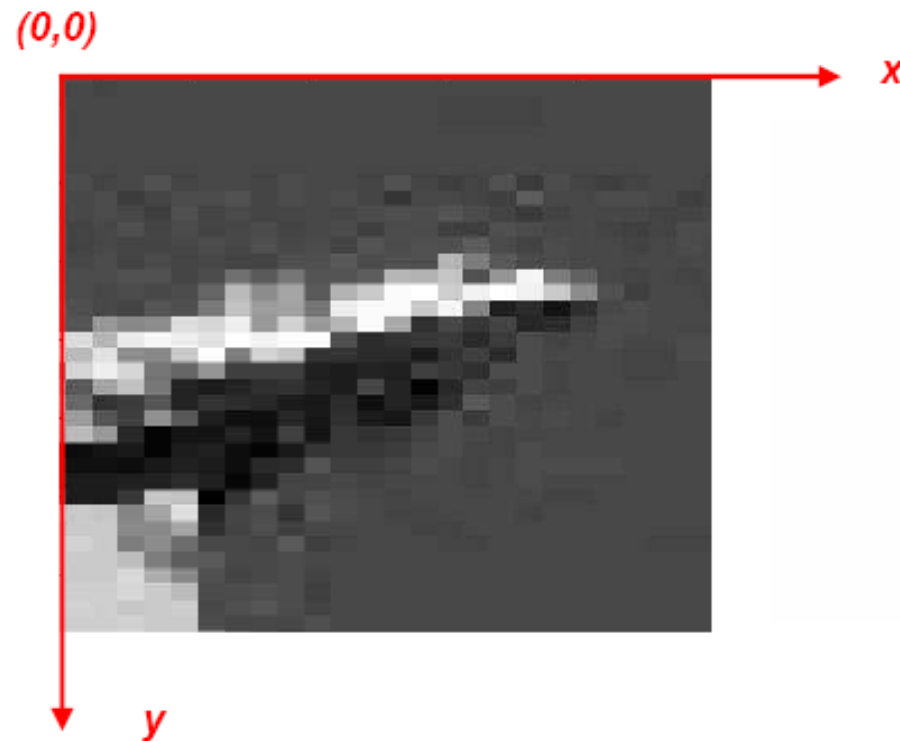
61	29	29	57	199	192
222	200	197	135	167	222
203	203	203	137	137	165
208	208	201	124	142	111
208	203	200	190	127	92
204	201	200	218	173	139

It's just a bunch of NUMBERS!

Pixel = "picture element"

- Represents brightness at one point

Digital Image Representation



- Images are contiguous blocks of numbers in computer memory
- We will manipulate these numbers to get them into a useful form

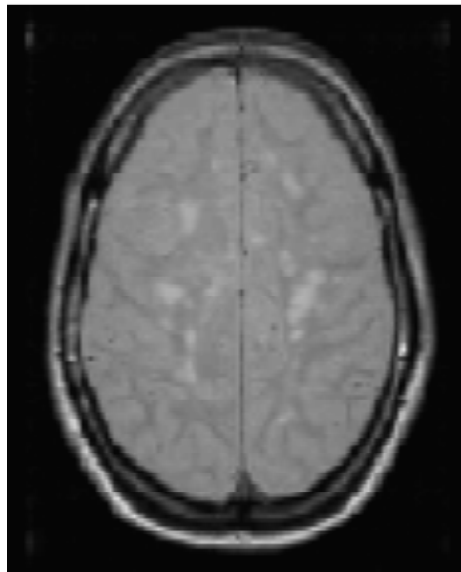
Pixel

Pixel value describes how bright that pixel is, and/or what color it should be.

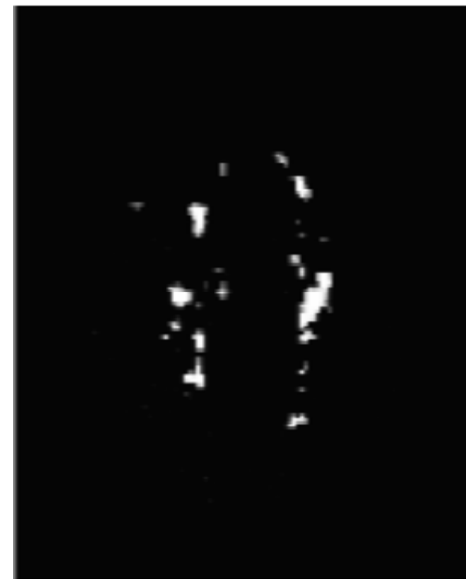
- **Binary images** – 1-bit number indicating foreground or background.
- **Gray scale images** – 8-bit number .
 - Typically 0= black, 255=white
 - Values between 0 and 255 = different shades of gray.



Binary image & Gray scale image



Gray-scale MRI image



Binary Image –
segmented MS
lesion sites

Color images

- Separate R-G-B components for each pixel
- Pixel values is actually a vector of three numbers

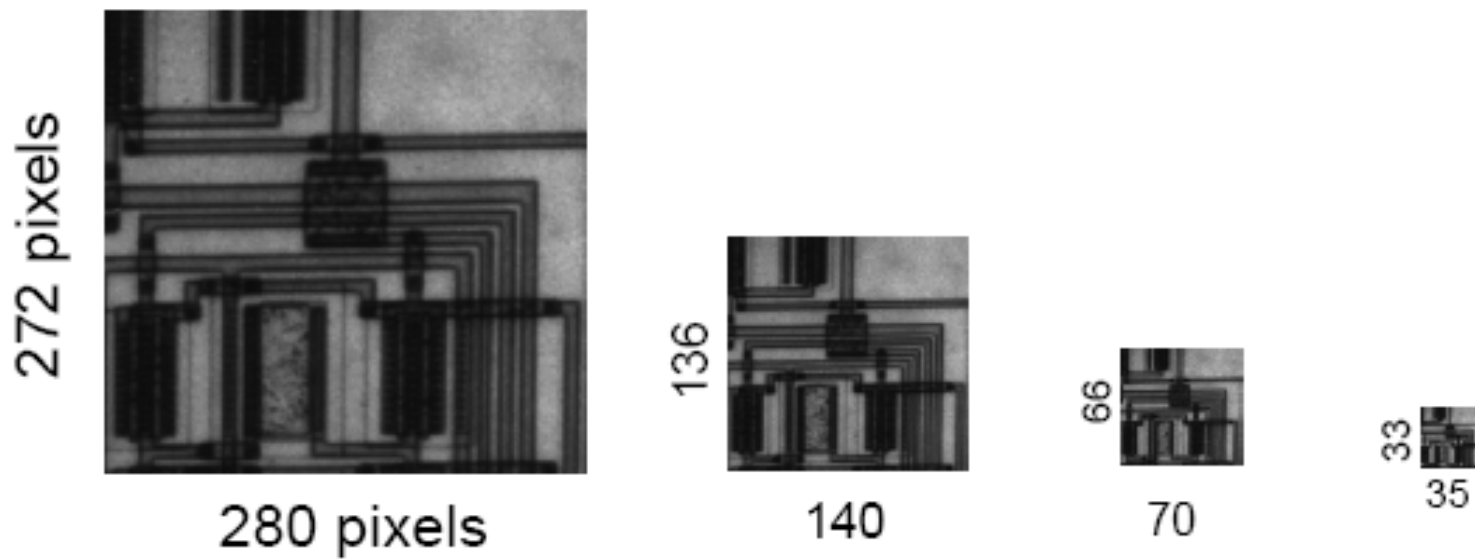


Contrast Resolution

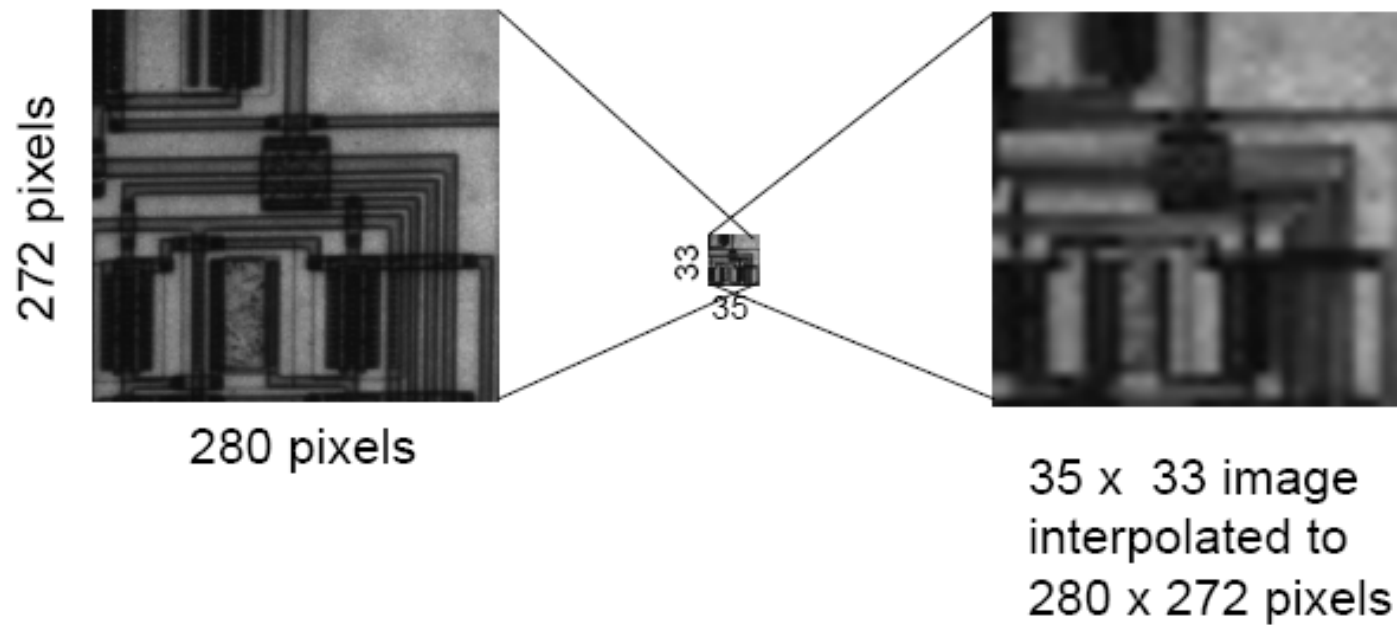


Different numbers of gray levels

Images of Different Sizes



Fewer Pixels Mean Lower Spatial Resolution



Properties that define the image format

- Pixel (or spatial) resolution (e.g. 640x480 pixels)
- Pixel bit-depth (8-bit unsigned, 16-bit signed, etc.)
- Number of planes - 1 for grayscale images, 3 for color
- Colorspace (RGB, YUV, etc.)
- Pixel store format (planar vs. packed)
 - In a packed format, the components are stored in a single array.
 - In a planar format, the components are stored separately, as three planes.

RGBRGBRGB.....RGB

RR.....RGG...GGB....B

Digital Image Representation (cont'd)

- A digital image is the representation of a continuous image $f(x,y)$ by a 2-d array of discrete samples.
- A pixel is an element of the 2-d array of samples.
- Pixels are point samples, without extent.

But the Real World is Continuous!

- So how do we go from a continuous physical world

... .. to a discrete image world?

The answer is

Sampling & Quantization!

Digital Image Acquisition Process

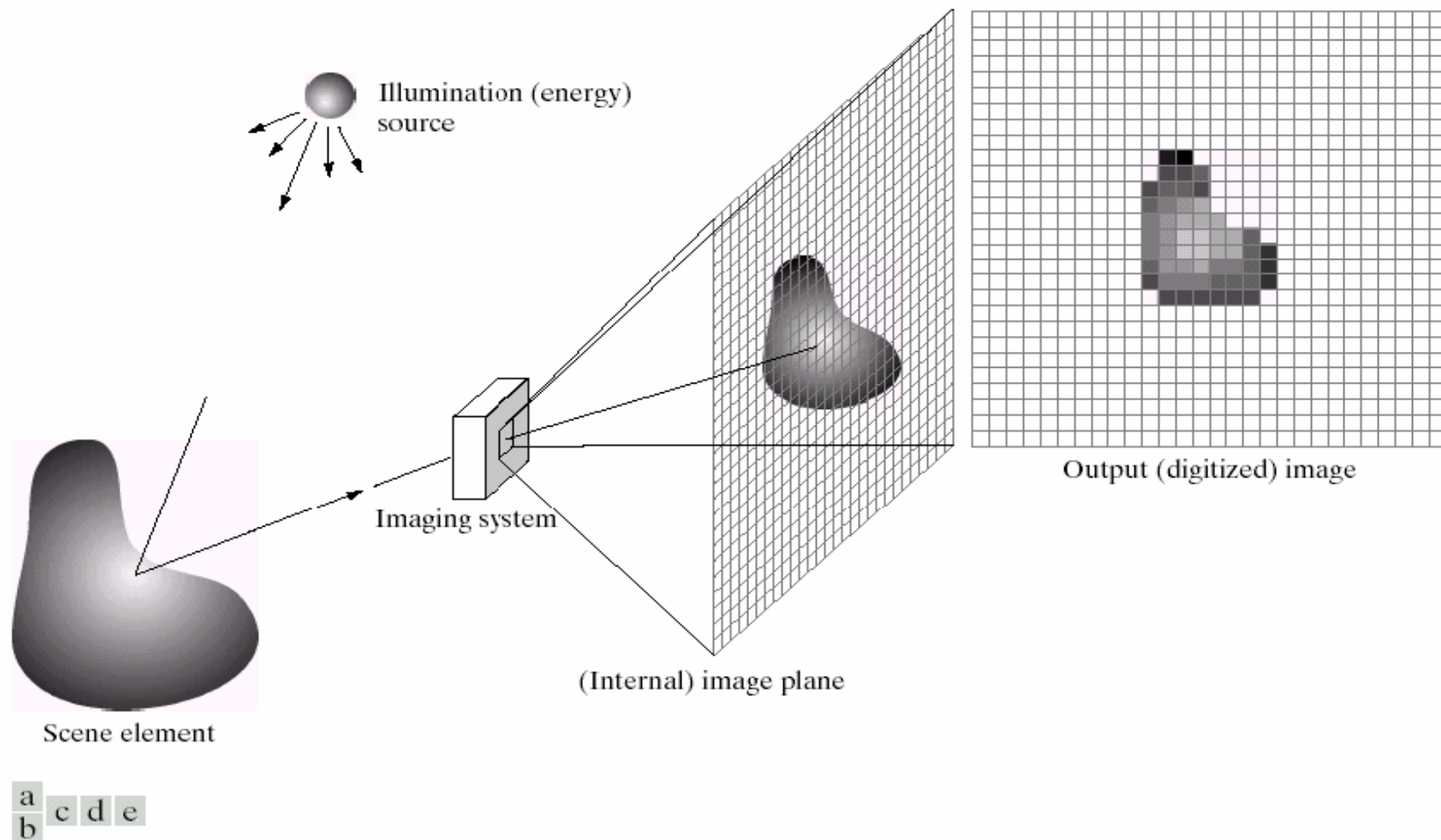
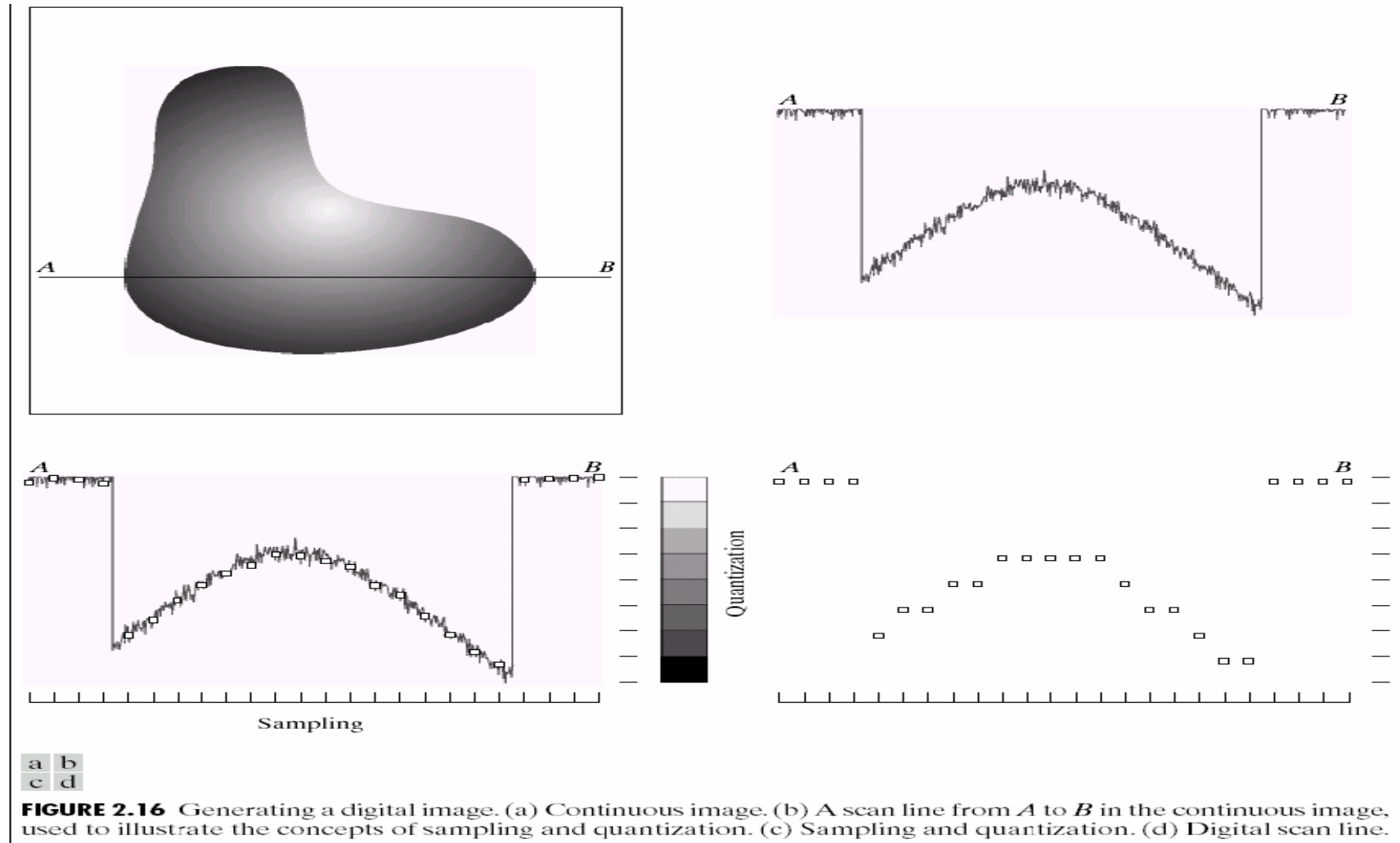
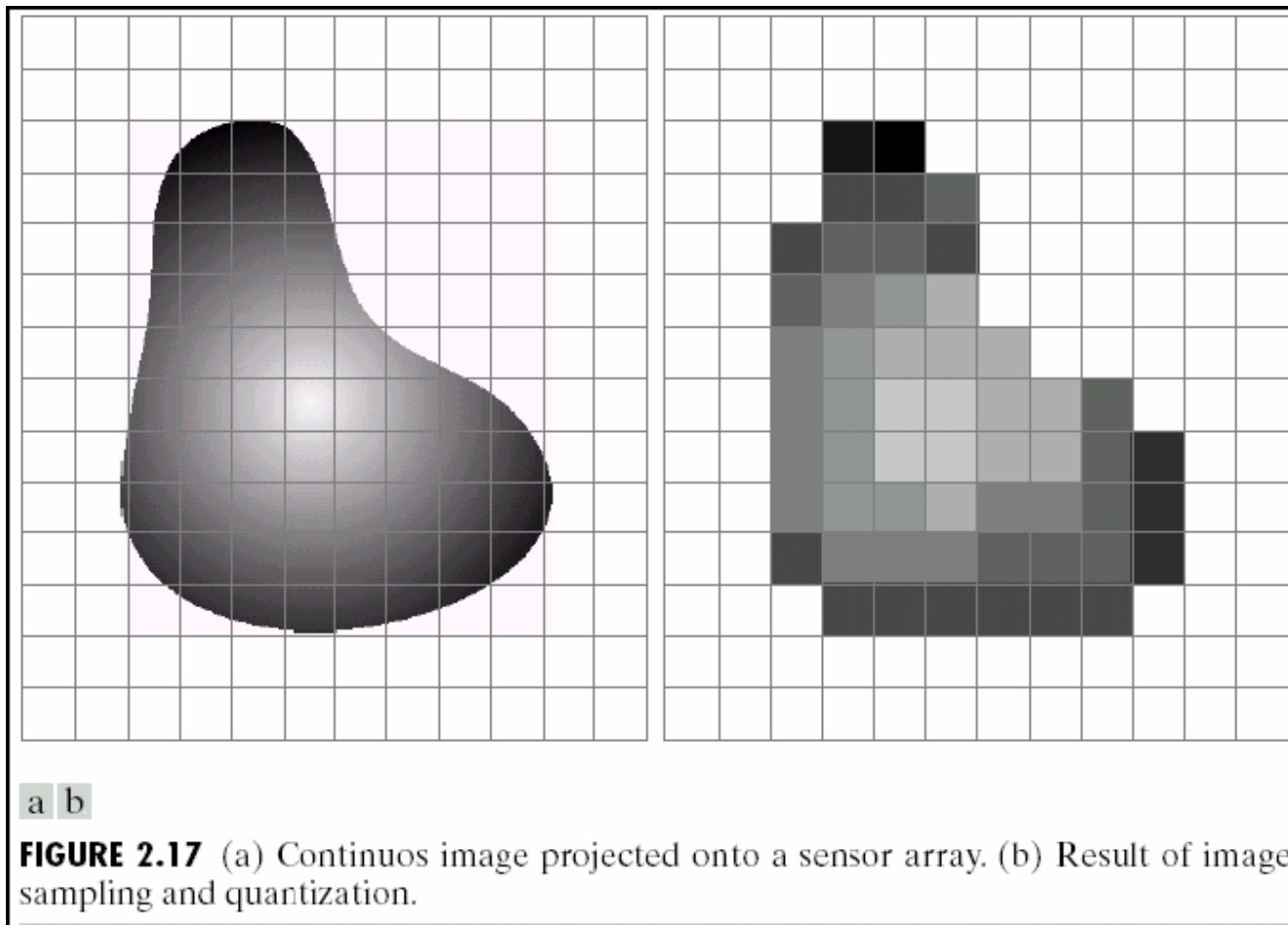


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy (“illumination”) source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Concepts of Sampling & Quantization

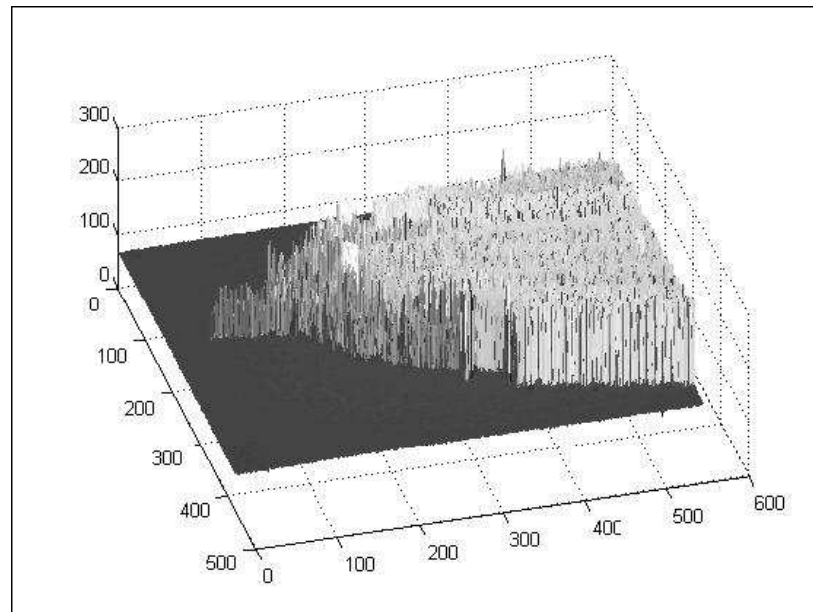


Concepts of Sampling & Quantization



Images are Discrete Functions

- correspond to the average scene luminance as perceived by the camera over a period of time.



Since Images are just Discrete Functions

- We can do image arithmetic

```
for(i = 0; i < numPixels; i++)  
    im3[i] = im1[i] + im2[i];
```

- We can do binary operations

```
for(i = 0; i < numPixels; i++)  
    im3[i] = im1[i] || im2[i];
```

etc.

Other image data

- **Video** is nothing more than images over time

$$I(x, y) \Rightarrow I(x, y, t)$$

- **Multi-Spectral images** – several images of the same scene, each of them taken with different sensors.
 - Can have more than three components for each pixel.
 - Eg – MRI images, remote sensing images

Assignment 1

- In Matlab, using the help information under the **image processing toolbox**
- Click on **Getting Started** and read/try out the following
 - **What Is the Image Processing Toolbox?**– introduction about the tool box
 - **Example 1** - guides you through an example of some of the basic image processing capabilities, including reading, writing, and displaying images
 - **Example 2** – guides you through some advanced image processing topics
 - **Getting Help** - provides pointers to additional sources of information

Questions....